Steffen **Plunder**

Mathematician · Developer · Interdisciplinary Researcher

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About Me

Computational researcher specializing in GPU-accelerated agent-based models coupled with PDEs for biological systems. Expertise in high-performance numerical methods and interdisciplinary bio-math research with proven track record of achieving 300× speedups through optimized algorithms and efficient implementation. Published in Nature Physics, Nature Materials, and Nature Communications. Strong foundation in numerical mathematics, nonsmooth analysis, and software engineering best practices.

Experience

Research Scientist, ASHBi, Kyoto University, Japan *02/2023 – present*

- Built GPU-accelerated simulations for digit growth and analyzed 3D image data (meshing, cell tracks, structural measures).
- Published in high impact journals Nature Physics, Nature Materials, and iScience.
- Initiated cross-department networking events to connect researchers across biology and mathematics. Co-supervised three students, co-organized international workshops.

Graduate Researcher, University of Vienna, Austria *10/2019 – 01/2023*

- Proved that a widely-used game engine technique (Position-based Dynamics) is scientifically accurate.
- Achieved 300× speedup in simulations through algorithm design and efficient implementation.
- Published two Nature Communications papers and 4 further manuscripts.

Research Assistant, TU Kaiserslautern, Germany *12/2018 – 10/2019*

- Organized calculus classes and exams. Research in Differential-Algebraic Systems group leading to 2 publications.

Programming Assistant, Fraunhofer Institute ITWM, Kaiserslautern, Germany *06/2013 – 06/2016*

- C++ programming for image analysis software and user interfaces (Qt).

Student Assistant, TU Kaiserslautern, Germany *06/2016 – 09/2017*

- Implementation of B-Spline based parameterisation algorithms (C++/G+SMo) at the differential-algebraic systems group of Prof. Simeon.

Education

German grading system: 1 = excellent, 4 = passing, 6 = fail.

Ph.D. in Mathematics, University of Vienna, Austria Grade: 1.0 *10/2019 – 01/2023*

- Supervised by Dr. Sara Merino-Aceituno
- Focus: Kinetic theory and applications to biology

M.Sc. in Mathematics, TU Kaiserslautern, Germany Grade: 1.1 *10/2016 – 09/2018*

- Specialization: Partial differential equations, Application subject: Physics
- Thesis: Fiber based Lagrangian Modelling and Simulation of Skeletal Muscle Tissue
- ERASMUS semester at TU Delft (WS 2016/2017)

B.Sc. in Mathematics, TU Kaiserslautern, Germany Grade: 1.1 *10/2013 – 09/2016*

- Specialization: Modelling and scientific computing, Application subject: Physics
- Thesis: Molecular Dynamical Simulation for Polymers

High School (Gymnasium), Trifels Gymnasium Annweiler, Germany Grade: 1.6 *09/2005 – 06/2013*

Skills

HPC & GPU Computing

Agent-based modeling · PDE-coupled systems · GPU acceleration (Julia/Python) · Parallel algorithms · Performance optimization · CUDA (via frameworks) · WebGPU · Numerical methods

Programming Languages

Julia · Python · C++ · JavaScript/TypeScript · MATLAB · Lua

Scientific Computing & ML

SciML (Julia) · numpy/scipy · numba · jax · pytorch · FEniCSx · vtk/ParaView

Development Tools

Git/GitHub · Linux · LaTeX/Typst · Qt

Languages

German	Native
English	Fluent
Chinese	Basic (A1-A2)

Activities

Open Source Packages

SpatialHashTables.jl · BoundedDegreeGraphs.jl · InkReveal · MyPaintedSlides

Past Engagement

ASHBi Bar (networking event) · VSM Student Speaker · Mentor for exchange students

Awards and Grants

Kakenhi: Early-Career Scientists (PI) ≈ 12,000€ — 04/2024 – 03/2027

ASHBi Fusion Grant (Co-PI) ≈ 30,000€ — 04/2023 – 03/2025

EMBO Scientific Exchange Grant 2,700€ — 01/2021

Award for Master's Thesis, Kreisparkassenstiftung Kaiserslautern — 07/2019

DAAD-PROMOS Scholarship — 01/2018

Main Award, DMV Students Conference 2017 — 01/2017

Felix-Klein Scholarship, Fraunhofer ITWM — 10/2016 – 09/2018

Deutschlandstipendium — 10/2014 – 09/2016

Teaching Experience

Lecture: Parameter Estimation for Differential Equations, Hokkaido University — *12/2024*

Short Programming Course: Julia, University of Vienna — *01/2022*

Short Programming Course: Python, Vienna School of Mathematics — *01/2022*

Organisation and Tutor, TU Kaiserslautern — *10/2018 – 09/2019*

Tutor, TU Kaiserslautern — *10/2014 – 09/2018*

- Numerical methods for linear algebra and analysis
- Höhere Mathematik I: Analysis (for engineers)
- Foundation of Mathematics I & II (for Mathematicians and Physicists)

Presentations

Modeling the Directionality of Cell Migration in EMT with Non-Smooth Dynamical Systems, NCTS Webinar on Nonlinear Evolutionary Dynamics *09/2025*

Position-based Dynamics for Multicellular Systems with Adhesion Memory, ACMB-JSMB 2025, Kyoto *07/2025*

Emergence of digit-like structures through mechanical instabilities, ASHBi Retreat 2025, Shiga *01/2025*

Agent-based modelling of heterogeneous EMT scenarios, Annual Meeting ESMTB, Toledo *07/2024*

Tools for agent-based modelling in developmental biology, JuliaCon 2024, Eindhoven *07/2024*

Position-based dynamics as a numerical tool for agent-based cell migration models, KSMB-SMB 2024, Seoul *07/2024*

Position-based dynamics: Convergence theory and applications to agent-based models, PDE seminar, Vienna *03/2024*

Convergence of position based dynamics for first-order particle systems, ICIAM 2023, Tokyo *08/2023*

Position-based dynamics for ODEs with inequality constraints, PDE afternoon, Vienna *11/2021*

Partially kinetic systems (aka 'particles on rails'), Kinetic Theory Coffee Break, online *05/2020*

Lagrangian perspective on skeletal muscle models, GAMM, 90th Annual Meeting, Vienna *02/2019*

Fiber based muscle simulation (master's thesis), Hausdorff Center of Mathematics, Bonn *10/2018*

Simulation of skeletal muscle tissue with fiber-based dynamics, Applied Math Seminar, Kaiserslautern *06/2018*

Symplectic numerical integration, PhD seminar, Auckland *04/2018*

Symplectic molecular dynamics, 19th ÖMG Meeting and Annual DMV Meeting, Salzburg *09/2017*

Organization and Outreach

Workshop Co-Organizer, Development across Scale 2023, Kyoto *01/2023*

Workshop Co-Organizer, Vienna School of Mathematics *01/2021 – 12/2022*

- VSM Mini-Course on String Theory by Pavel Safronov (one week)
- VSM workshop: A PhD in mathematics – career possibilities & gender aspects (one day)

Outreach Activity, Lange Nacht der Wissenschaft, Vienna *01/2022*

Student Talks Initiator, TU Kaiserslautern *01/2017 – 12/2019*

Research Stays

Academic Visit, Paul Sabatier University, Toulouse, France *01/2021*

- Collaboration with Dr. Eric Theveneau on heterogeneity of cell dynamics and tissue mechanics during epithelial-to-mesenchymal transitions (EMT)
- Six weeks stay funded by EMBO Scientific Exchange Grant

Academic Visit, Hausdorff Center of Mathematics, Bonn, Germany *01/2018*

- Extension of multiphase material simulation
- Supervised by Prof. Martin Rumpf and Dr. Behrend Heeren

Academic Visit, University of Auckland, New Zealand *01/2018*

- Multiscale simulation of flows in saliva glands

- Supervised by Prof. James Sneyd
- Funded by DAAD-PROMOS scholarship

Selected Projects

SpatialHashTables.jl

- GPU-accelerated spatial hashing for fast collision detection in unbounded domains
- Supports both CPU and GPU backends through Julia's generic programming

BoundedDegreeGraphs.jl

- Allocation-free operations optimized for agent-based modeling
- GPU-compatible design through Julia's array abstraction frameworks

MyPaintEdSlides

- Combining hand-drawn aesthetics with digital flexibility

Publications

Total: 12 peer-reviewed publications, 2 preprints

P. C. Guruciaga, T. Ichikawa, **S. Plunder**, T. Hiiragi, A. Erzberger, *Boundary geometry controls a topological defect transition that determines lumen nucleation in embryonic development*, **Nature Materials (2026)**

T. Ichikawa, P. C. Guruciaga, S. Hu, **S. Plunder**, M. Makino, M. Hamaji, A. Stokkermans, S. Yoshida, A. Erzberger, T. Hiiragi, *Boundary-guided cell alignment drives mouse epiblast maturation*, **Nature Physics (2026)**

R. Tsutsumi, A. Diez, **S. Plunder**, R. Kimura, S. Oki, K. Takizawa, H. Akiyama, Y. Mii, R. Takada, S. Takada, M. Eiraku, *Self-organized fingering instabilities drive the emergence of tissue morphogenesis in digit organoids*, **bioRxiv preprint (2025)**

M. Mira Osuna, **S. Plunder**, E. Theveneau, R. Le Borgne, *The directionality of collective cell delamination is governed by tissue architecture and cell adhesion in a Drosophila carcinoma model*, **iScience (2025)**

S. Merino-Aceituno, **S. Plunder**, C. Wytrzens, H. Yoldaş, *Macroscopic effects of an anisotropic Gaussian-type repulsive potential: nematic alignment and spatial effects*, **Mathematical Models and Methods in Applied Sciences (2025)**

S. Plunder, C. Danesin, B. Glise, M. A. Ferreira, S. Merino-Aceituno, E. Theveneau, *Modelling variability and heterogeneity of EMT scenarios highlights nuclear positioning and protrusions as main drivers of extrusion*, **Nature Communications (2024)**

E. Despin-Guitard, V. S. Rosa, **S. Plunder**, N. Mathiah, K. Van Schoor, E. Nehme, S. Merino-Aceituno, J. Egea, M. N. Shahbazi, E. Theveneau, I. Migeotte, *Non-apical mitoses contribute to cell delamination during mouse gastrulation*, **Nature Communications (2024)**

S. Plunder, M. Burkard, T. Helling, U. M. Lauer, L. E. Hoelzle, L. Marongiu, *Determination of optimal phage load and administration time for antibacterial treatment*, **Current Protocols (2024)**

H. Jäger, É. Grosjean, **S. Plunder**, C. Redenbach, A. Keilmann, B. Simeon, C. Surulescu, *Cell seeding dynamics in a porous scaffold material designed for meniscus tissue regeneration*, **PAMM (2024)**

S. Plunder, S. Merino-Aceituno, *Convergence proof for first-order position-based dynamics: An efficient scheme for inequality constrained ODEs*, **arXiv preprint (2023)**

S. Plunder, B. Simeon, *The mean-field limit for particle systems with uniform full-rank constraints*, **Kinetic and Related Models (2023)**

S. Plunder, M. Burkard, U. Lauer, S. Venturelli, L. Marongiu, *Determination of phage load and administration time in simulated occurrences of antibacterial treatments*, **Frontiers in Medicine (2022)**

S. Plunder, B. Simeon, *Coupled Systems of Linear Differential-Algebraic and Kinetic Equations with Application to the Mathematical Modelling of Muscle Tissue*, **Progress in Differential-Algebraic Equations II (2020)**